

MINAS KATSIOKALIS

XR Software Engineer

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SUMMARY

I am an eXtended Reality (XR) Software Engineer. I have experience on both handheld and head-mounted display (HMD) XR applications, and I continuously explore the vast potential of XR technology. I have years of hands-on experience in Unity and .NET/C# framework, always investigating smart solutions and optimized methods. I am familiar with agile mythologies and tools. My goal is to contribute to the advancement of immersive technologies and uncover innovative and fun ways to enhance users' experience.

SKILLS

Languages: C#/.NET, Python, JavaScript, HTML/CSS, C/C++
Engines: Unity3D
XR SDKs: AR Foundation (ARCore/ARKit), Vuforia, MRTK (OpenXR, OculusXR), Meta XR SDK, Lightship ARDK
Platforms: Quest 2/3/Pro, Hololens2, Vision Pro, Android, iOS
Versioning: Git

EXPERIENCE

- 8/2024 – Present **XR Software Engineer | Full Time** **ANIMA RES GmbH**
- Contribute to the development pipeline of mobile and HMD applications across multiple platforms, including Apple Vision Pro, HoloLens, Android, and iOS.
 - Implement new features and enhancements for the company's Insight app gallery, improving functionality and user experience.
 - Collaborate closely with 3D and technical artists to ensure high-quality visuals and optimal performance.
 - Maintain, troubleshoot, and upgrade the company's custom-built package system to support ongoing development needs.
 - Meet deadlines for publishing updates and new releases of applications (e.g. Insight Heart) ensuring alignment with project timelines and quality standards.
- Unity3D / C# / AR Foundation / MRTK / Android(ARCore)/iOS(ARKit) / Polyspatial
- 10/2021 – 7/2024 **AR Software Engineer | Full Time** **Up2metric P.C.**
- Design & Implementation of Mobile AR applications (Android/iOS).
 - Troubleshooting and problem-solving of AR related issues.
 - Integration of REST API calls into front-end interfaces.
 - Responsible for prototyping, building and maintaining AR features.
 - Research on XR standards and specifications.
- Unity3D / C# / AR Foundation / Vuforia / Android(ARCore)/iOS(ARKit) / Figma
- 9/2021 – 3/2023 **XR Software Engineer & Researcher | Full Time** **Technical University of Crete**
- Design & Implementation of XR HMD applications (Hololens2/Meta Quest).
 - Research on XR Collaborative environments, and multimodal communication techniques.
 - Prototyping and testing on AR HMD features.
 - Publication and attendance on international conferences in the field of XR.
 - Paper reviewer in IEEE VR 2023 & 2024.
 - Participation in HORIZON EU Research Projects (SUN XR, VARCITIES).
- Unity3D / C# / MRTK / Meta XR SDK / Vuforia / HCI
- 2022,2023 Spring Semesters **Graphics Lab Assistant | Part Time** **Technical University of Crete**
- As post-graduate student I contributed to my supervisor's courses:
- **Graphics Course (Undergraduate - 4th year):**
 - Delivered practical instruction in Unity3D for desktop game development.
 - Designed, prepared, and tested semester-long student projects using Unity3D.
 - **Human-Computer Interaction (HCI) Course (Postgraduate):**
 - Organized and guided the semester project, including use-case design, persona development, usability evaluation, UX design, prototyping, and web app implementation.
 - Contributed to the preparation, review, and evaluation of course materials and assignments.
- Unity3D / C# / HCI
- 9/2018 – 5/2020 **AR Researcher & Developer | Part Time** **Technical University of Crete**
Diploma Thesis
- Develop a Mobile AR application for Android devices.
 - Integrate navigation system using Mapbox SDK.
 - Research and testing on various AR SDKs: Vuforia, Wikitude, ARCore, ARKit etc.
 - Research on the field of Mobile AR (MAR).
- Unity3D / C# / AR Foundation / Vuforia / ARCore / Wikitude

EDUCATION

2020 **MEng, Integrated Masters in Electrical and Computer Engineering**
(5 Year Course - 300 ECTS) **Technical University of Crete**
Diploma Thesis: "Outdoors Mobile Augmented Reality for Coastal Erosion Visualization, based on Geographical Data" a Mobile AR application for Android devices, developed in Unity3D game engine.

PUBLICATIONS

11/2024 **International Conference on Extended Reality, XR Salento 2024** **Springer**
Engineering a BIM-Based Mixed Reality Application for the Life-Cycle Management of Buildings

12/2023 **Journal: Smart Cities 2024, 7(1), 1-32** **MDPI**
The Intersection of the Green and the Smart City: A Data Platform for Health and Well-Being through Nature-Based Solutions

6/2023 **ACM International Conference on Interactive Media Experiences (IMX' 23)** **ACM Library**
GoNature AR: Air Quality & Noise Visualization Through a Multimodal and Interactive Augmented Reality Experience

6/2022 **7th International Conference on Smart and Sustainable Technologies (SpliTech)** **IEEE Xplore**
On the integration of nature-based solutions with digital innovation for health and wellbeing in cities

11/2020 **International Workshop on Cross-Reality (XR) Interaction @ ACM ISS 2020** **Workshop Proceedings**
Outdoors Mobile Augmented Reality for Coastal Erosion Visualization, based on Geographical Data

* *Google Scholar.*

AWARDS

6/2023 **Best Demo Award** **ACM International Conference on Interactive Media Experiences (IMX' 23)**
Received "Best Demo Award" of the conference by Tobbi for the demo: "GoNature AR".

2/2022 **Winner of "Innovation" Category** **Brackeys GameJam 2023.1**
Category winners for the game: "UnSentenced"

LANGUAGES

• **English** - Fluent • **German** - Beginner • **Greek** - Native